# Mahmoud Alnabawi

**Data Analyst | Quality Analytics | Business Intelligence**

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## Professional Summary

Data Analyst and manufacturing quality professional with experience in laboratory testing, calibration tracking, supplier compliance, and quality reporting. Uses Excel, Power Query, Power BI, DAX, SQL, and Python to organize data, monitor KPIs, identify trends, and support operational decisions. Applies Root Cause Analysis, Pareto Analysis, and Lean Six Sigma to investigate quality deviations. Combines hands-on manufacturing experience with formal training in data analytics and data science.

## Technical Skills

**Data Analysis:** Data Cleaning, Data Validation, Exploratory Data Analysis (EDA), Trend Analysis, Descriptive Statistics, KPI Analysis, Data Quality Checks
**Business Intelligence:** Power BI, DAX, Power Query, Data Modeling, Dashboard Development, Data Visualization, Reporting
**Excel:** Pivot Tables, Lookup Functions, Data Cleaning, Reporting, VBA Automation
**SQL and Python:** SQL (SELECT, JOIN, GROUP BY, Subqueries, Aggregations), Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
**Quality and Compliance:** Root Cause Analysis (RCA), Pareto Analysis, Lean Six Sigma, DMAIC, Defect Analysis, ISO 9001, ISO 14001, ISO 45001, ISO/IEC 17025

## Professional Experience

**QA & Sustainability Specialist | Raw Material and External Laboratory Testing**
**Mac Carpet – Oriental Weavers Group, Egypt** **Jul 2024 – Present**

- Managed calibration tracking for approximately **700 measuring and testing devices**, maintaining records and monitoring due dates to support equipment traceability and quality control.
- Developed an Excel-based external laboratory tracking system to follow sample dispatch, laboratory receipt, report delivery, invoicing, and payment status.
- Monitored approximately **20 external laboratory samples per month** and maintained tracking records over a **three-month monitoring period**, improving visibility of pending tests and follow-up actions.
- Coordinated testing and consolidated analytical results from **3 external laboratories: Intertek, SGS, and Bureau Veritas**, supporting material evaluation, supplier compliance, and quality decisions.
- Supported packaging compliance activities related to the EU Packaging and Packaging Waste Regulation (PPWR) and PFAS, including heavy-metal checks, supplier packaging declarations, and an action plan covering **6 material categories**.
- Applied Root Cause Analysis, Pareto Analysis, and Lean Six Sigma methods to investigate quality deviations, document corrective actions, and support KPI reporting.

**Quality Control Engineer**
**Extrucoat – Aluminum Extrusion and Coating, Egypt** **Nov 2022 – Jul 2024**

- Performed laboratory and production-line quality checks on aluminum billets and profiles, including hardness testing and curing-temperature monitoring.
- Monitored chemical pretreatment and wastewater-treatment processes, recorded inspection results, and reported deviations for corrective action.
- Maintained quality records and supported ISO 9001 documentation, product conformity checks, and production-quality monitoring.

## Data Analytics Projects

**Manufacturing KPI Dashboard | Power BI**

- Transformed manufacturing and quality data using Power Query and developed DAX measures to monitor KPIs and analyze trends.
- Designed interactive dashboard views to support performance monitoring and data-driven discussion of manufacturing and quality indicators.

**Manufacturing Quality and Defect Analysis | Excel**

- Cleaned and aggregated quality records, then used trend and Pareto analysis to identify major contributors to

recorded defects.
- Organized findings to support root-cause investigation and corrective-action prioritization.

**Diabetes Risk Dataset Audit | Data Analytics Project**

- Audited a **41-column dataset** as part of a **five-member team**, assessed missing values, performed data cleaning, and analyzed the prepared data.
- Documented data-quality issues and presented findings to support the team's analytical workflow.

## Education

**Bachelor of Science in Applied Chemistry** **2022**
Faculty of Science, Al-Azhar University, Egypt

## Certifications and Professional Development

- IBM Data Science Professional Certificate – Coursera
- Google Advanced Data Analytics Certificate – Coursera
- Lean Six Sigma Black Belt
- Digital Egypt Pioneers Initiative (DEPI), Data Analytics for Professionals – In Progress
- Training in ISO 9001, ISO 14001, ISO 45001, and ISO/IEC 17025

## Additional Information

**Content Creation:** Created Arabic-language educational video content on Python and SQL fundamentals.
**Languages:** Arabic (Native); English (add your accurate proficiency level before applying).